Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · simplest form *(āhua māmā rawa)*

A A fraction of an amount ●○○

Remember: Divide by the denominator, then multiply by the numerator: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{2}$ of 18 = _____ **b)** $\frac{1}{4}$ of 20 = _____ **c)** $\frac{1}{3}$ of 27 = _____ **d)** $\frac{1}{5}$ of 35 = _____

e) $\frac{3}{4}$ of 20 = _____ **f)** $\frac{2}{3}$ of 15 = _____ **g)** $\frac{2}{5}$ of 30 = _____ **h)** $\frac{5}{6}$ of 24 = _____

i) $\frac{3}{8}$ of 40 = _____ **j)** $\frac{7}{10}$ of 60 = _____ **k)** $\frac{4}{9}$ of 36 = _____ **l)** $\frac{3}{5}$ of 100 = _____

B A whole number times a fraction ●●○

Watch out: Multiply the numerator only: $3 \times \frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$. The denominator does not change.

2. Work out. Give each answer in its simplest form, as a mixed number if it is more than 1.

a) $\frac{1}{5} \times 3$ **b)** $\frac{2}{7} \times 3$ **c)** $\frac{3}{4} \times 5$ **d)** $\frac{2}{3} \times 4$

e) $\frac{5}{6} \times 3$ **f)** $\frac{3}{8} \times 4$ **g)** $\frac{7}{10} \times 5$ **h)** $\frac{4}{9} \times 6$

i) $\frac{1}{2} \times 7$ **j)** $\frac{3}{5} \times 10$ **k)** $\frac{2}{9} \times 3$ **l)** $\frac{5}{12} \times 8$

C Finding the whole ●●●

Remember: Work back to one part first. If $\frac{3}{4}$ of a number is 12, then $\frac{1}{4}$ of it is $12 \div 3 = 4$, so the number is $4 \times 4 = 16$.

3. Find the number.

a) $\frac{1}{2}$ of a number is 9 **b)** $\frac{1}{3}$ of a number is 7 **c)** $\frac{1}{5}$ of a number is 9

d) $\frac{3}{4}$ of a number is 12 **e)** $\frac{2}{3}$ of a number is 10 **f)** $\frac{2}{5}$ of a number is 8

g) $\frac{5}{6}$ of a number is 20 **h)** $\frac{3}{10}$ of a number is 21 **i)** $\frac{7}{8}$ of a number is 35

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D A fraction divided by a whole number ●●●

> **Remember:** Dividing a fraction by 2 makes each piece half as big, so the denominator doubles: $\frac{1}{3} \div 2 = \frac{1}{6}$.

4. Work out. Give each answer in its simplest form.

a) $\frac{1}{2} \div 3$ **b)** $\frac{1}{4} \div 2$ **c)** $\frac{2}{3} \div 2$ **d)** $\frac{3}{4} \div 3$

e) $\frac{1}{5} \div 4$ **f)** $\frac{4}{5} \div 2$ **g)** $\frac{5}{6} \div 5$ **h)** $\frac{3}{8} \div 2$

E A whole number divided by a unit fraction ●●●

> **Remember:** $3 \div \frac{1}{4}$ asks "how many quarters are there in 3?" There are 4 in each whole, so the answer is $3 \times 4 = 12$.

5. Work out.

a) $2 \div \frac{1}{2}$ = _____ **b)** $3 \div \frac{1}{4}$ = _____ **c)** $5 \div \frac{1}{3}$ = _____ **d)** $4 \div \frac{1}{5}$ = _____

e) $6 \div \frac{1}{2}$ = _____ **f)** $2 \div \frac{1}{10}$ = _____ **g)** $7 \div \frac{1}{4}$ = _____ **h)** $10 \div \frac{1}{3}$ = _____

F Problems ●●●

6. A class has 28 students. $\frac{3}{4}$ of them have a pet. How many students have a pet?

7. A ribbon is 3 m long. It is cut into pieces that are each $\frac{1}{4}$ m long. How many pieces are there?

8. Mele has read $\frac{2}{5}$ of her book. She has read 14 pages. How many pages does the book have?

9. $\frac{1}{2}$ of a pizza is shared equally between 4 people. What fraction of the whole pizza does each person get?

★ Extension

10. Find $\frac{2}{3}$ of 10. The answer is not a whole number.

11. Work out $6 \div \frac{1}{4}$, then explain why the answer is bigger than 6.

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Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · simplest form *(āhua māmā rawa)*

A A fraction of an amount ●○○

Remember: Divide by the denominator, then multiply by the numerator: $\frac{3}{4}$ of 20 is $20 \div 4 \times 3 = 15$.

1. Work out.

a) $\frac{1}{2}$ of 26 = _____ **b)** $\frac{1}{4}$ of 28 = _____ **c)** $\frac{1}{3}$ of 21 = _____ **d)** $\frac{1}{5}$ of 45 = _____

e) $\frac{3}{4}$ of 16 = _____ **f)** $\frac{2}{3}$ of 18 = _____ **g)** $\frac{3}{5}$ of 20 = _____ **h)** $\frac{5}{6}$ of 30 = _____

i) $\frac{3}{8}$ of 32 = _____ **j)** $\frac{9}{10}$ of 50 = _____ **k)** $\frac{4}{7}$ of 35 = _____ **l)** $\frac{2}{5}$ of 100 = _____

B A whole number times a fraction ●●○

Watch out: Multiply the numerator only: $3 \times \frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$. The denominator does not change.

2. Work out. Give each answer in its simplest form, as a mixed number if it is more than 1.

a) $\frac{1}{7} \times 4$ **b)** $\frac{2}{9} \times 4$ **c)** $\frac{3}{4} \times 7$ **d)** $\frac{2}{3} \times 5$

e) $\frac{5}{6} \times 4$ **f)** $\frac{5}{8} \times 4$ **g)** $\frac{3}{10} \times 5$ **h)** $\frac{2}{9} \times 6$

i) $\frac{1}{2} \times 9$ **j)** $\frac{4}{5} \times 10$ **k)** $\frac{4}{9} \times 3$ **l)** $\frac{7}{12} \times 8$

C Finding the whole ●●●

Remember: Work back to one part first. If $\frac{3}{4}$ of a number is 12, then $\frac{1}{4}$ of it is $12 \div 3 = 4$, so the number is $4 \times 4 = 16$.

3. Find the number.

a) $\frac{1}{2}$ of a number is 13 **b)** $\frac{1}{4}$ of a number is 6 **c)** $\frac{1}{6}$ of a number is 8

d) $\frac{3}{4}$ of a number is 15 **e)** $\frac{2}{3}$ of a number is 14 **f)** $\frac{2}{5}$ of a number is 12

g) $\frac{5}{6}$ of a number is 25 **h)** $\frac{3}{10}$ of a number is 27 **i)** $\frac{5}{8}$ of a number is 45

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D A fraction divided by a whole number ●●●

Remember: Dividing a fraction by 2 makes each piece half as big, so the denominator doubles: $\frac{1}{3} \div 2 = \frac{1}{6}$.

4. Work out. Give each answer in its simplest form.

a) $\frac{1}{3} \div 2$ **b)** $\frac{1}{2} \div 5$ **c)** $\frac{4}{5} \div 4$ **d)** $\frac{2}{3} \div 4$

e) $\frac{1}{4} \div 3$ **f)** $\frac{6}{7} \div 3$ **g)** $\frac{3}{4} \div 6$ **h)** $\frac{5}{8} \div 2$

E A whole number divided by a unit fraction ●●●

Remember: $3 \div \frac{1}{4}$ asks "how many quarters are there in 3?" There are 4 in each whole, so the answer is $3 \times 4 = 12$.

5. Work out.

a) $3 \div \frac{1}{2}$ = _____ **b)** $2 \div \frac{1}{4}$ = _____ **c)** $4 \div \frac{1}{3}$ = _____ **d)** $3 \div \frac{1}{5}$ = _____

e) $8 \div \frac{1}{2}$ = _____ **f)** $3 \div \frac{1}{10}$ = _____ **g)** $5 \div \frac{1}{4}$ = _____ **h)** $10 \div \frac{1}{5}$ = _____

F Problems ●●●

6. A class has 27 students. $\frac{2}{3}$ of them have a pet. How many students have a pet?

7. A ribbon is 4 m long. It is cut into pieces that are each $\frac{1}{3}$ m long. How many pieces are there?

8. Mele has read $\frac{3}{5}$ of her book. She has read 24 pages. How many pages does the book have?

9. $\frac{1}{2}$ of a pizza is shared equally between 3 people. What fraction of the whole pizza does each person get?

★ Extension

10. Find $\frac{3}{4}$ of 9. The answer is not a whole number.

11. Work out $8 \div \frac{1}{3}$, then explain why the answer is bigger than 8.

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